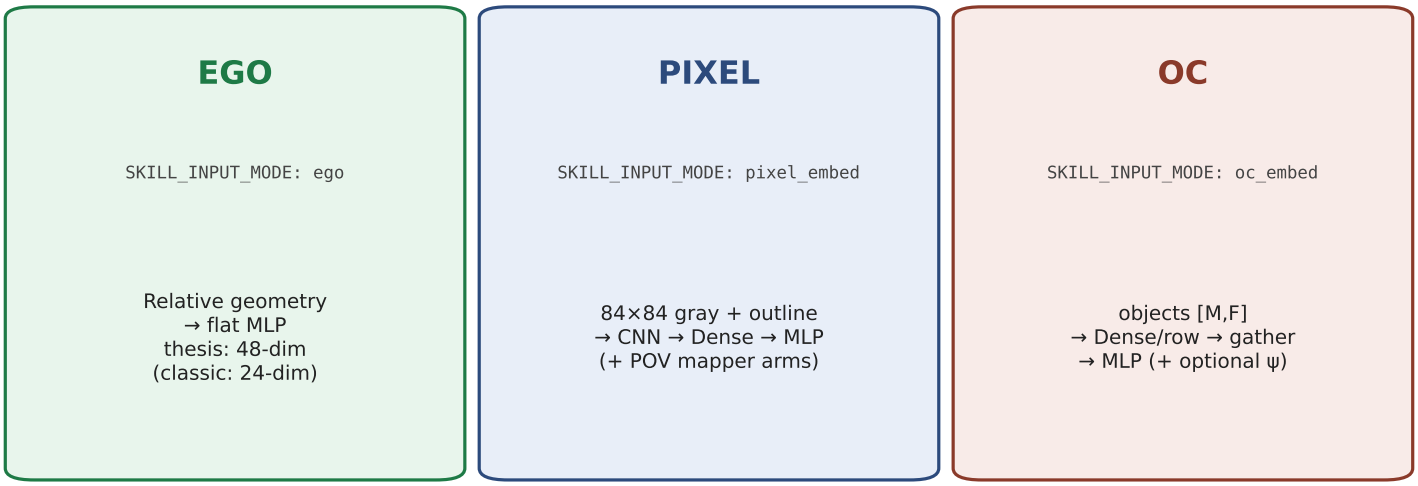
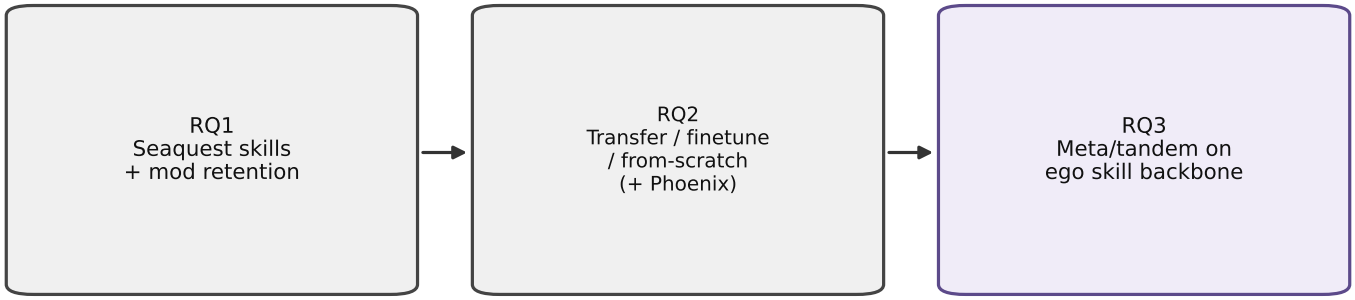
Paper Eval Architecture

Concept-Based Learning — System-1 observation modes + suite protocol

Current thesis spine: POV facing-frame ego · φ8 bins · clip50 → 48-dim @ stack 4



Shared rule: target_idx is a gather key — never a slot one-hot

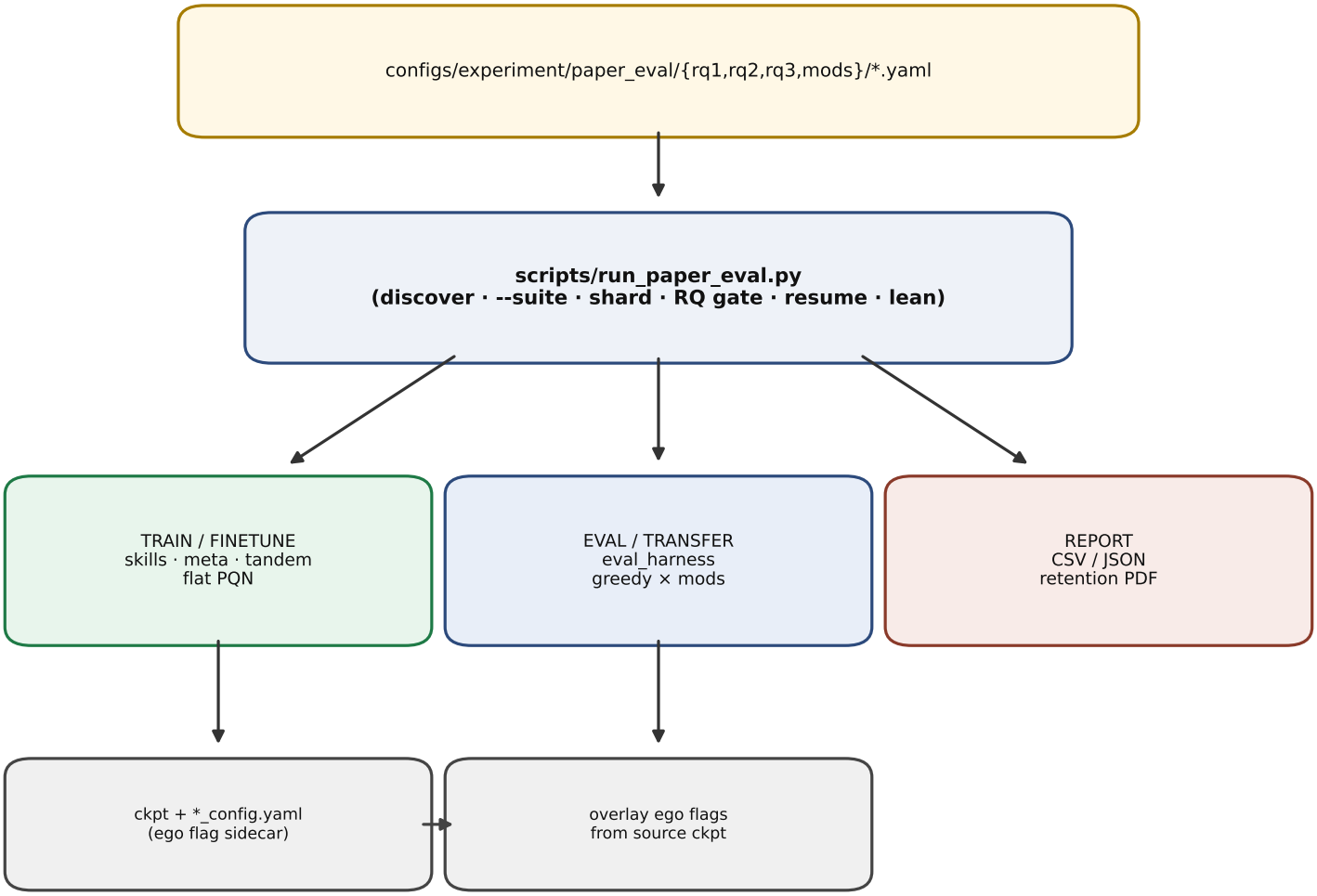


docs/skill_input_modes.md · configs/experiment/paper_eval/ · scripts/run_paper_eval.py

Regenerate: python scripts/build_paper_eval_architecture_pdf.py

1. Suite pipeline (diagram)

How configs become train / eval / reports



Matched outer loop (RQ1 skill arms)

TOTAL_TIMESTEPS ≈ 24.98M · NUM_ENVS=128 · NUM_STEPS=128 · FRAME_SKIP=4 · stack=4

Skills: DESTROY / COLLECT / AVOID / NAVIGATE · MAX_OBJECTS=35 · RELABEL_ALL_OBJECTS=false

Ego encoding flags copied onto pixel/OC YAMLs for sidecar consistency (even if unused by those nets).

RQ dependencies + named suites

RQ2/RQ3 wait on RQ1 Seaquest skill ckpts (PAPER_EVAL_REQUIRES_RQ1 / --wait-deps).

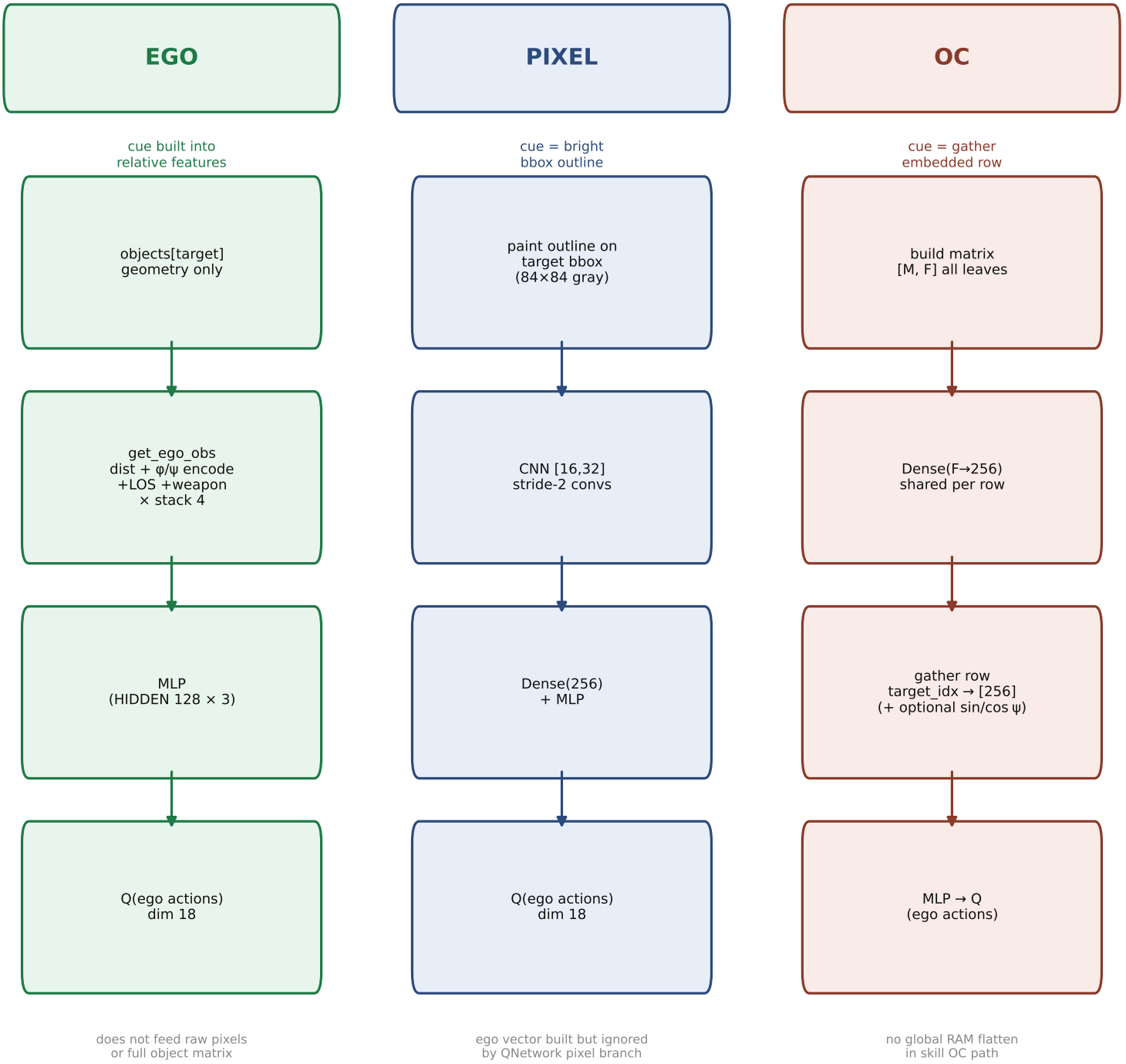
Named suites (thesis_facing, pov_facing_v2, clip50_angbins_full, …) select stems; not only --rq.

RQ3 skills are always ego; meta input is pixel_embed or oc_embed.

2. Three System-1 architectures (side-by-side)

Same outer loop; different observation + target cue. target_idx never enters as a one-hot.

pixel_obs | objects (labeled) | target_idx (gather key)



Where target_idx is used

EGO: select geometry for feature builder only

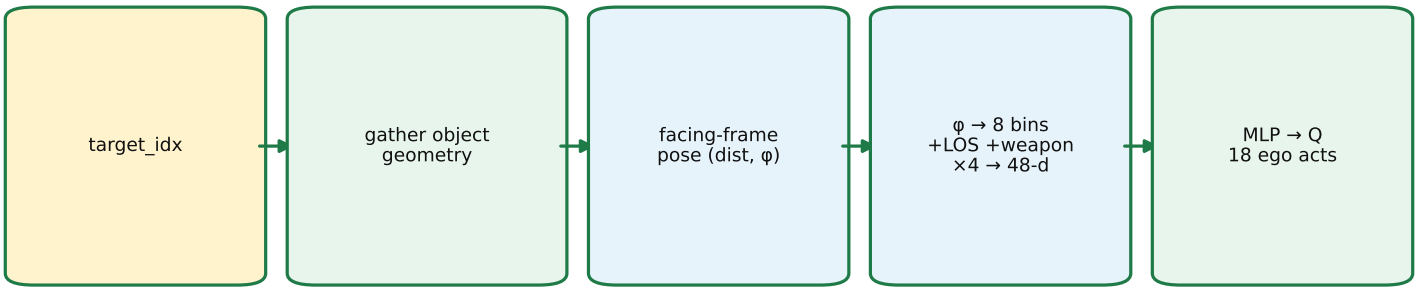
PIXEL: select bbox to paint (outline in pixel space)

OC: gather which row of the embedded object matrix

NEVER: one_hot(target_idx, MAX_OBJECTS) concatenated into the net (removed — broke remapping)

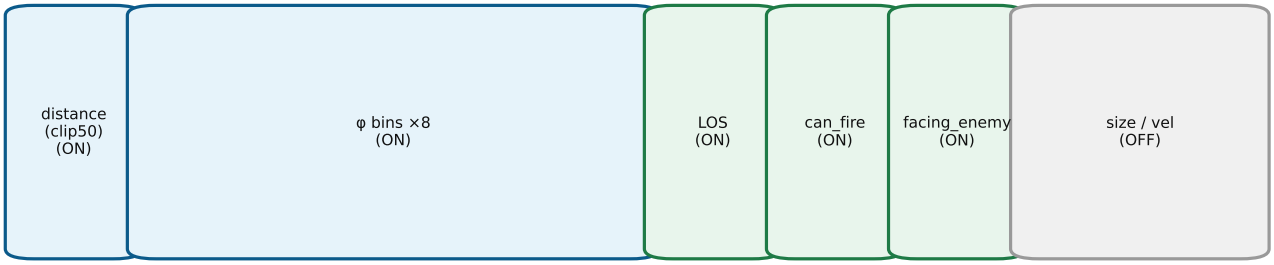
3. Ego architecture (detail)

Thesis spine vs classic baseline — layout must match the Seaquest ckpt sidecar on transfer



Thesis spine: USE_POV_FACING_FRAME + EGO_PSI_ANGLE_BINS=8 + clip50

Thesis spine channels / frame (×4 stack = 48)



φ = bearing of target in player-facing body frame | SIZE/VELOCITY always off for paper arms

Encoding families (transfer locks to the trained layout)

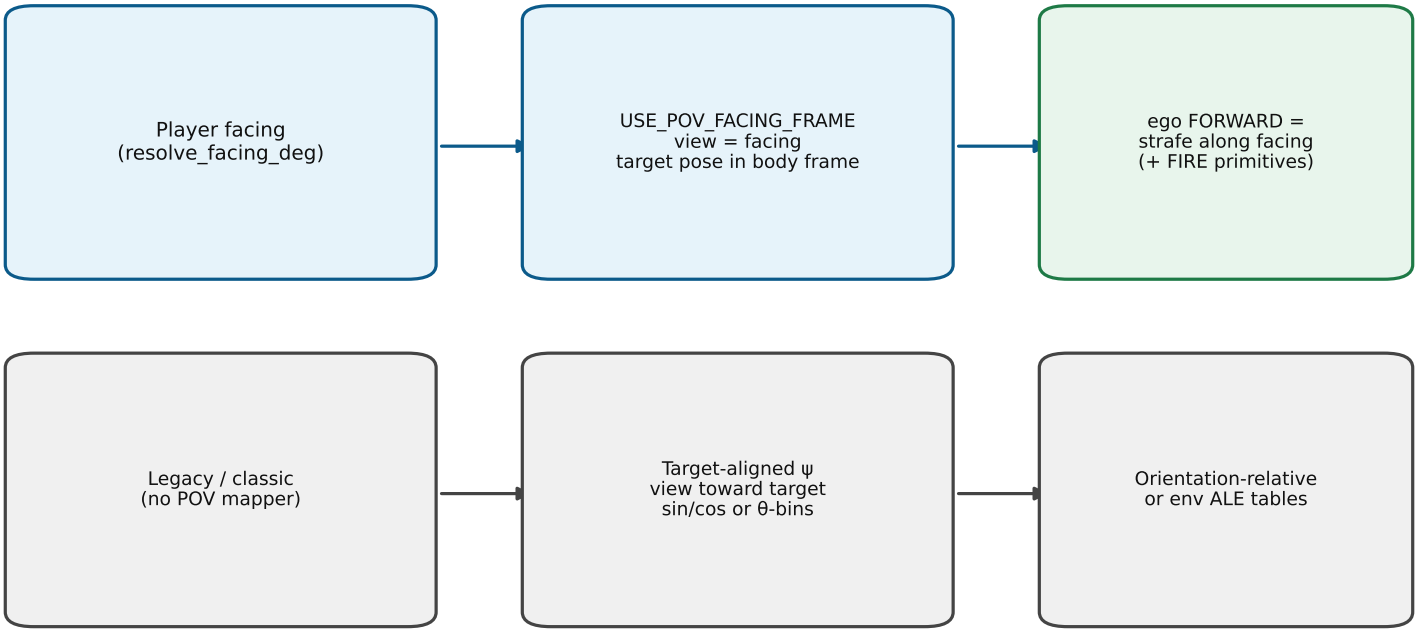
Thesis / facing spine: POV facing frame · EGO_PSI_ANGLE_BINS=8 · clip=50 → 12 ch/frame → 48-dim
Classic baseline: target-aligned sin/cos(ψ) · blocking+weapon · clip null/50 → 6 ch/frame → 24-dim
World-θ bins (non-POV): EGO_ANGLE_BINS=8 · clip50 → also 48-dim (different angle semantics)
Ablations: clip0 / clip100 / distflags / facing_xy / restrained pose / shot extras — see YAML stems

Shared flags

SIZE=false · VELOCITY=false · BLOCKING=true · WEAPON=true · USE_LEGACY_EGO_OBS=false
USE_LEGACY_SEAQUEST_* action/ascent/death guards = false · EGO_ACTION_DIM=18
Pre-2026-08-30 facing-frame ckpts invalid (obs/mapper fix) — use *_v2 / thesis_facing stems.

4. POV action mapper + ego frame

Body-relative control must match observation semantics



Flags

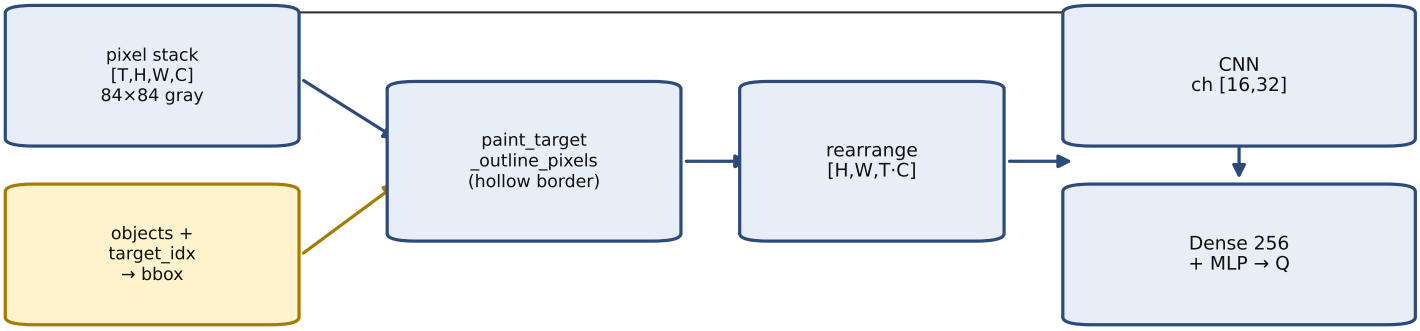
USE_POV_RELATIVE_MAPPER — body-facing ego action table (required for facing / fixed2 / turn).
USE_POV_FACING_FRAME — FPS obs: φ from player facing (thesis spine).
EGO_POSE_XY — emit (fwd, lat) instead of (distance, φ) under facing frame.
USE_POV_FIXED_BILATERAL / USE_POV_TURN_PRIMITIVES — alternate POV ablations (not thesis default).

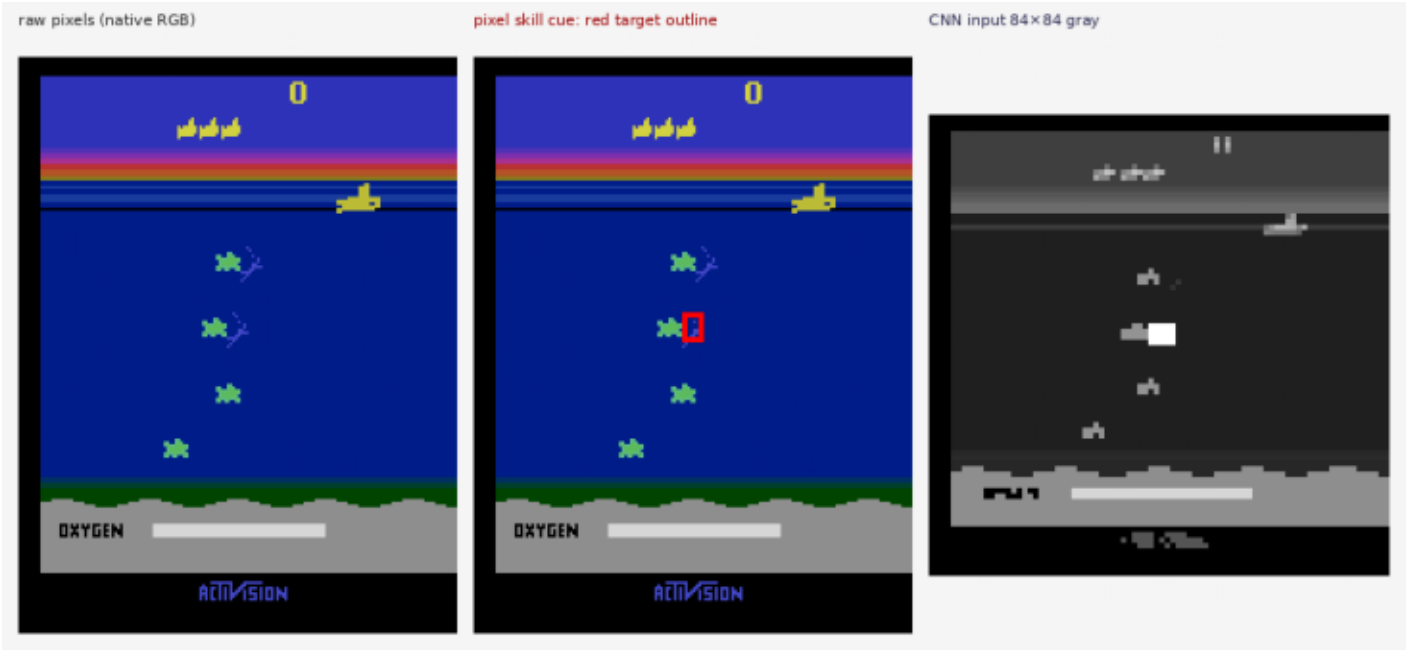
Pixel / OC under POV

thesis_facing pairs pixel_pov / oc_pov arms with the same facing-relative mapper so
System-1 action semantics match across observation modes (obs cue still outline / gather).
Skill OC may set SKILL_OC_PSI_FEATURES to concat sin/cos(ψ) after gather (facing − bearing).

5. Pixel architecture (detail)

SKILL_INPUT_MODE: pixel_embed — CNN + painted target outline





Seaquest skill input: raw RGB → red target outline cue → 84×84 grayscale CNN frame

Settings: META_PIXEL_DO_RESIZE=true · RESIZE [84,84] · GRAYSCALE=true · NATIVE_DOWNSCALING=true · SKILL_PIXEL_CNN_CHANNELS=[16,32]

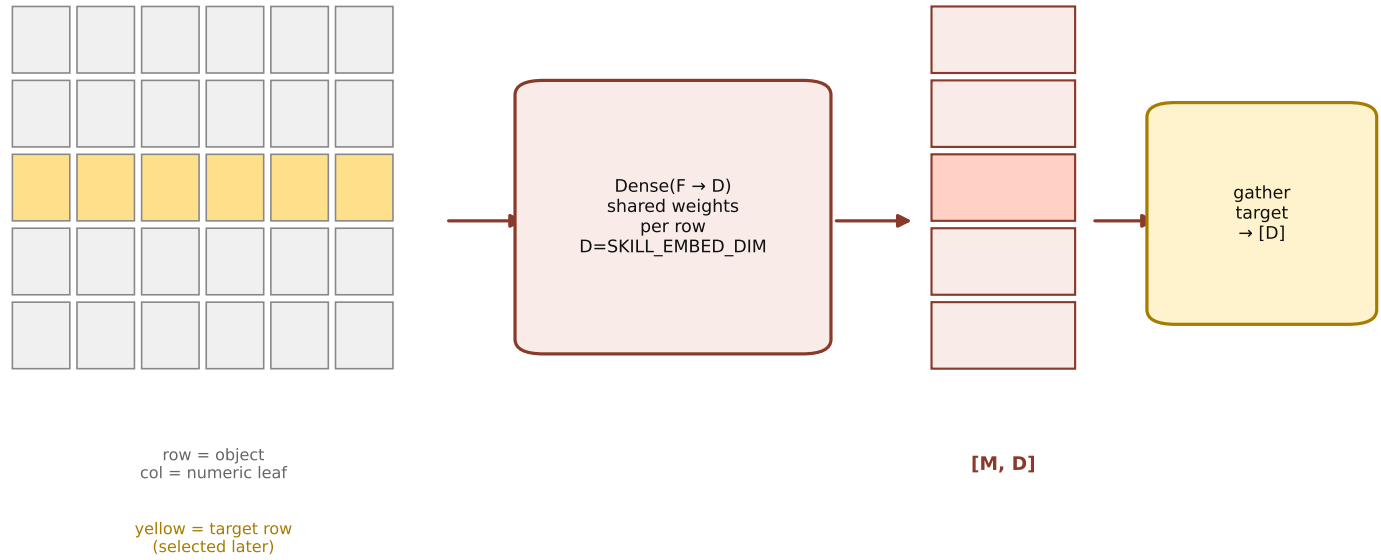
Design notes

- Outline replaces slot one-hot / tiled slot embed (those failed remapping ablations).
- Same ~25M env-step budget as ego despite ~10× wall time (RTPT weight ≈ 10).
- Ablation: SKILL_PIXEL_PAINT_TARGET_OUTLINE=false (nooutline) — transfer should collapse.
- Flat PQN (seaquest_pqn) is a separate non-hierarchical pixel baseline at same budget.
- RQ2 from-scratch includes pixel; RQ3 uses pixel as META_INPUT_MODE over ego skills.
- thesis_facing uses *_pixel_pov stems (POV mapper; outline cue unchanged).
- Screenshot asset: docs/paper_eval/pixel_skill_target_outline.png

6. OC architecture (detail)

SKILL_INPUT_MODE: oc_embed — per-object Dense + gather (not flat M·F)

objects → matrix [M, F]



KEPT: Dense(F→D) + gather
• kernel width = F only
• M can differ across games
• loads when leaf schema shared

REMOVED: Dense(M·F → D)
• first-layer width ∝ M
• breaks cross-game load
• + old one_hot(target) cue

Leaves in F (same source meta uses per candidate)

bbox, active, visual_id, state, orientation, semantic_label, flags, …
Skill OC does NOT concat a global raw RAM flatten.

Meta vs skill OC · POV ψ

META_INPUT_MODE: oc_embed concatenates full object matrix onto a global flatten —
no transferable per-object Dense there. That Dense lives on skill oc_embed only.
SKILL_OC_PSI_FEATURES: after gather, concat sin/cos(ψ) (facing − bearing) for POV arms.

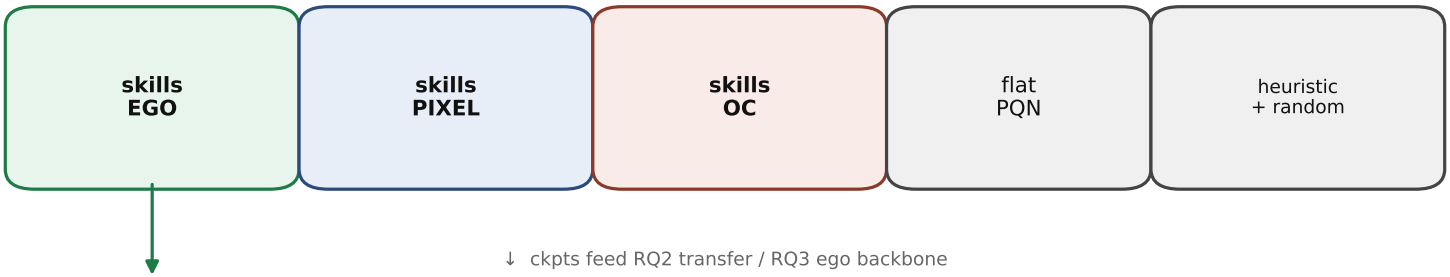
Suite role

RQ1 train · RQ2 zero-shot + finetune · NOT from-scratch (protocol: ego+pixel only).
RQ3: oc_embed as meta input over ego skills. RTPT weight ≈ 2×.

7. RQ protocol map

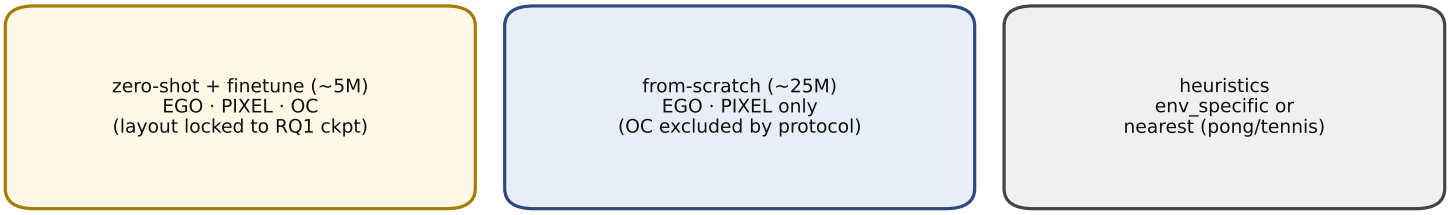
Which observation mode is used where (thesis_facing + classic paper_eval)

RQ1 — Seaquest (in-distribution + mod eval)

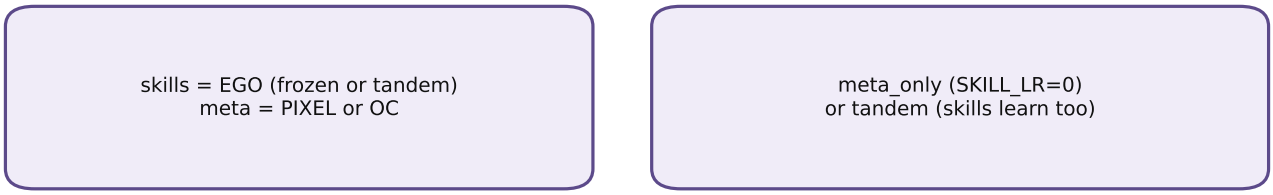


↓ ckpts feed RQ2 transfer / RQ3 ego backbone

RQ2 — pong · tennis · beamrider · freeway · asterix · phoenix



RQ3 — System 2 on Seaquest



Cross-cutting

- Mod allowlists: configs/experiment/paper_eval/mods/ (control / semantic / dynamics / visual).
- Eval overlays ego encoding flags from ckpt sidecar (clip / bins / POV / weapon / blocking).
- RTPT weights: ego×1, oc×2, pixel×10 (wall-time calibration, not step budget).
- Reporting: env_metrics beyond score; retention = return_mod / return_base.
- Active suite names: thesis_facing · pov_facing_v2 · clip50_angbins_full · … (run_paper_eval.py).

Authoritative sources

docs/skill_input_modes.md · paper_eval YAML headers · observations/egocentric.py
observations/target_conditioning.py · networks/skill_network.py · paper_eval/eval_harness.py

8. Text reference (flags & decisions)

Quick lookup — diagrams above are the primary overview

Ego — thesis spine (48-dim)

POV_FACING_FRAME · PSI_ANGLE_BINS=8 · clip=50 · SIZE/VEL=false · BLOCKING/WEAPON=true

Stem family: *_facing_psibins8_* / thesis_facing · transfer must match this layout.

Ego — classic baseline (24-dim)

sin/cos(ψ) · blocking+weapon · no POV facing · SIZE/VEL=false · clip often null or 50

Pixel

pixel_embed · 84×84 gray · CNN [16,32] · outline on target bbox · optional *_pov mapper

Index-invariant visual cue; matched step budget; ~10× wall time.

OC

oc_embed · [M,F] → Dense(F→D)/row → gather(target) → MLP · optional SKILL_OC_PSI_FEATURES

No skill RAM flatten; no from-scratch in RQ2; meta OC ≠ skill OC Dense.

Removed

one_hot(target_idx) · tiled slot embed in CNN · flat Dense(M·F→D)

Shared RQ1 loop

4 skills · 25M steps · 128×128 · skip 4 · stack 4 · MAX_OBJECTS 35 · relabel off

SKILL_DEATH_PENALIZE_COLLECT=true · HEURISTIC_MODE=env_specific (Seaquest)

